

Math 147

Analysis

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2 Foundations of Calculus

2.1 Lecture 1

2.1.1 What Is Calculus?

Differential and integral calculus are two branches of calculus.

Differential calculus: The study of slopes and tangent lines.

Integral calculus: The study of areas and accumulation.

2.1.2 Numbers

The number systems discussed in this lecture include

$$\begin{aligned}\mathbb{N} &= \{1, 2, 3, \dots\}, \\ \mathbb{Z} &= \{\dots, -1, 0, 1, \dots\}, \\ \mathbb{Q} &= \left\{ \frac{m}{n} : m, n \in \mathbb{Z}, n \neq 0 \right\}.\end{aligned}$$

The natural numbers are closed under addition. The integers are closed under addition, subtraction, and multiplication, but they are not closed under division; for example, $1/2 \notin \mathbb{Z}$. Therefore, we introduce the rational numbers to include quotients of integers with nonzero denominators.

Theorem 2.1.1 (Irrationality of $\sqrt{2}$)

The number $\sqrt{2}$ is irrational; equivalently, $\sqrt{2} \notin \mathbb{Q}$.

Proof

Suppose, for contradiction, that $\sqrt{2} \in \mathbb{Q}$. Then we can write

$$\sqrt{2} = \frac{m}{n}, \quad m, n \in \mathbb{Z}, \quad n > 0, \quad \gcd(m, n) = 1.$$

Squaring both sides gives

$$2 = \frac{m^2}{n^2} \quad \implies \quad m^2 = 2n^2.$$

Thus, m^2 is even, so m is even. Write $m = 2\ell$ for some $\ell \in \mathbb{Z}$. Substituting this into the equation above gives

$$\begin{aligned}2n^2 &= 4\ell^2, \\ n^2 &= 2\ell^2.\end{aligned}$$

Therefore, n^2 is even, and hence n is even as well.

This means that both m and n are even, contradicting $\gcd(m, n) = 1$. Therefore, $\sqrt{2} \notin \mathbb{Q}$. ■

The irrational numbers are the real numbers that are not rational:

$$\mathbb{R} \setminus \mathbb{Q} = \{x \in \mathbb{R} : x \notin \mathbb{Q}\}.$$

Example 2.1.2 (Decimal representation)

Every real number has a decimal representation. In general, a decimal expansion can be written as

$$x = a + \sum_{i=1}^{\infty} \frac{x_i}{10^i}, \quad a \in \mathbb{Z}, \quad x_i \in \{0, 1, 2, \dots, 9\}.$$

Example 2.1.3 (Recurring decimal)

Is the statement $1.999 \dots = 2$ correct?

Yes. Let $x = 1.999 \dots$. Then

$$\begin{aligned} x &= 1.999 \dots, \\ 10x &= 19.999 \dots, \\ 9x &= 18, \\ x &= 2. \end{aligned}$$

2.1.3 Statements and Quantifiers

Definition 2.1.4 (Statement)

A statement is a sentence that has a definite truth value: it is either true or false.

The **universal quantifier** $\forall$ means “for all.” For example,

Example 2.1.5

The statement

$$\forall x \in \mathbb{R}, \quad x^2 - x \geq 0$$

is false; for instance, it fails when $x = \frac{1}{2}$.

The **existential quantifier** $\exists$ means “there exists.” For example,

Example 2.1.6

The statement

$$\exists x \in \mathbb{R}, \quad x^2 - x \geq 0$$

is true; for instance, $x = 2$ satisfies the inequality.

Proposition 2.1.7

For every $x \in \mathbb{R}$, there exists $y \in \mathbb{R}$ such that

$$x^3 - y^3 = 1.$$

Proof

Let $x \in \mathbb{R}$ and take $y = \sqrt[3]{x^3 - 1}$. Then

$$x^3 - y^3 = x^3 - (x^3 - 1) = 1.$$

■

Proposition 2.1.8

The statement

$$\exists y \in \mathbb{R}, \quad \forall x \in \mathbb{R}, \quad x^3 - y^3 = 1$$

is false.

Its negation is

$$\forall y \in \mathbb{R}, \quad \exists x \in \mathbb{R}, \quad x^3 - y^3 \neq 1.$$

To prove the negation, let $y \in \mathbb{R}$ and take $x = y$. Then $x^3 - y^3 = 0 \neq 1$.

Remark

The order of quantifiers matters.

2.2 Mathematical Induction

Definition 2.2.1 (Principle of Mathematical Induction)

Let $P(n)$ be a statement depending on $n \in \mathbb{N}$. If the following two statements are true:

- (1) $P(1)$ is true;
- (2) for every $k \in \mathbb{N}$, if $P(k)$ is true, then $P(k + 1)$ is true;

then

$$\forall n \in \mathbb{N}, \quad P(n)$$

is true.

Here is an example:

Proposition 2.2.2 (Sum of the First n Natural Numbers)

For every $n \in \mathbb{N}$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof

Define $P(n)$ to be the statement

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

The base case is $n = 1$:

$$1 = \frac{1(1+1)}{2}.$$

For the inductive step, let $k \in \mathbb{N}$ and assume that $P(k)$ is true. Then

$$\begin{aligned} 1 + 2 + \cdots + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Thus, $P(k+1)$ is true. By the principle of mathematical induction, the proposition follows. ■

The principle of mathematical induction has the following equivalent set formulation.

Proposition 2.2.3

Let $S \subseteq \mathbb{N}$. If

- (i) $1 \in S$;
- (ii) for every $k \in S$, $k+1 \in S$;

then $S = \mathbb{N}$.

Proof

Define $P(n)$ to be the statement $n \in S$. The two assumptions give the base case $P(1)$ and the inductive implication $P(k) \Rightarrow P(k+1)$. Hence, by mathematical induction, every $n \in \mathbb{N}$ belongs to S . Since $S \subseteq \mathbb{N}$, it follows that $S = \mathbb{N}$. ■

2.2.1 Well-Ordering Principle**Theorem 2.2.4** (Well-Ordering Principle)

Every nonempty subset S of $\mathbb{N}$ has a smallest element.

Proof

Let $S \subseteq \mathbb{N}$ be nonempty. Suppose, for contradiction, that S has no smallest element. Define

$$T = \{n \in \mathbb{N} : \{1, 2, \dots, n\} \cap S = \emptyset\}.$$

Since S has no smallest element, $1 \notin S$, so $1 \in T$. Suppose $k \in T$. If $k + 1 \notin T$, then $k + 1 \in S$, while no element of $\{1, 2, \dots, k\}$ belongs to S . Thus, $k + 1$ would be the smallest element of S , a contradiction. Therefore, $k + 1 \in T$.

By mathematical induction, $T = \mathbb{N}$. This implies that no natural number belongs to S , so $S = \emptyset$, contradicting the assumption that S is nonempty. Therefore, S has a smallest element. ■

Proposition 2.2.5

Using the well-ordering principle, prove that for every $n \in \mathbb{N}$,

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Proof

Let $P(n)$ denote the statement

$$1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}.$$

Suppose, for contradiction, that $P(n)$ fails for some n ; that is,

$$\neg(\forall n \in \mathbb{N}, P(n)) \quad \text{equivalently} \quad \exists n \in \mathbb{N}, \neg P(n).$$

Then the set

$$S := \{n \in \mathbb{N} : \neg P(n)\}$$

is nonempty. By the well-ordering principle, S has a smallest element k . Since $P(1)$ holds, $1 \notin S$, so $k > 1$ and hence $k - 1 \in \mathbb{N}$.

By minimality of k , we have $k - 1 \notin S$, so $P(k - 1)$ is true:

$$1^2 + 2^2 + \dots + (k - 1)^2 = \frac{(k - 1)k(2(k - 1) + 1)}{6} = \frac{(k - 1)k(2k - 1)}{6}.$$

Adding k^2 to both sides gives

$$\begin{aligned} 1^2 + \dots + (k - 1)^2 + k^2 &= \frac{(k - 1)k(2k - 1)}{6} + k^2 \\ &= \frac{(k - 1)k(2k - 1) + 6k^2}{6} \\ &= \frac{k(k + 1)(2k + 1)}{6}, \end{aligned}$$

so $P(k)$ is in fact true. This contradicts $k \in S$. Therefore $S = \emptyset$, and $P(n)$ holds for every $n \in \mathbb{N}$. ■

2.3 The Algebraic and Order Properties of $\mathbb{R}$

The real numbers $\mathbb{R}$ form a *complete ordered field*. We build up this description in three stages: the field (algebraic) axioms, the order axioms, and the completeness axiom.

Definition 2.3.1 (Field)

A **field** is a set F equipped with two binary operations

$$+ : F \times F \rightarrow F, \quad \cdot : F \times F \rightarrow F,$$

satisfying the following axioms for all $a, b, c \in F$.

F1. Associativity of addition.

$$a + (b + c) = (a + b) + c.$$

F2. Commutativity of addition.

$$a + b = b + a.$$

F3. Additive identity.

There exists an element $0 \in F$ such that

$$a + 0 = a.$$

F4. Additive inverses.

For every $a \in F$, there exists an element $-a \in F$ such that

$$a + (-a) = 0.$$

F5. Associativity of multiplication.

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c.$$

F6. Commutativity of multiplication.

$$a \cdot b = b \cdot a.$$

F7. Multiplicative identity.

There exists an element $1 \in F$, with $1 \neq 0$, such that

$$a \cdot 1 = a.$$

F8. Multiplicative inverses.

For every $a \in F$ with $a \neq 0$, there exists an element $a^{-1} \in F$ such that

$$a \cdot a^{-1} = 1.$$

F9. Distributive law.

$$a \cdot (b + c) = a \cdot b + a \cdot c.$$

Remark

The rational numbers $\mathbb{Q}$ also form a field.

Theorem 2.3.2 a) If $z, a \in \mathbb{R}$ with $z + a = a$, then $z = 0$.
 b) If $u \in \mathbb{R}$ and $b \neq 0$ with $u \cdot b = b$, then $u = 1$.
 c) If $a \in \mathbb{R}$, then $a \cdot 0 = 0$.

Proof

a)

$$z = z + 0 = z + (a - a) = (z + a) + (-a) = a + (-a) = 0.$$

b) Multiplying $u \cdot b = b$ by b^{-1} ,

$$u = u \cdot 1 = u \cdot (b \cdot b^{-1}) = (u \cdot b) \cdot b^{-1} = b \cdot b^{-1} = 1.$$

c)

$$a \cdot 0 = a \cdot (1 + (-1)) = a \cdot 1 + a \cdot (-1) = a + (-a) = 0.$$

■

Definition 2.3.3 (The Order Properties of $\mathbb{R}$)

There is a nonempty subset P of $\mathbb{R}$, called the set of **positive real numbers**, that satisfies the following properties.

F11. Trichotomy law. For every $a \in \mathbb{R}$, exactly one of the following holds:

$$a = 0, \quad a \in P, \quad -a \in P.$$

F12. Closure under addition. If $a, b \in P$, then $a + b \in P$.

F13. Closure under multiplication. If $a, b \in P$, then $a \cdot b \in P$.

Definition 2.3.4 (Order relation)

Let $a, b \in \mathbb{R}$.

- (i) If $a - b \in P$, then we write $a > b$ or $b < a$.
- (ii) If $a - b \in P \cup \{0\}$, then we write $a \geq b$ or $b \leq a$.

Remark

$\mathbb{Q}$ is also ordered, so we are still missing something: completeness.

Definition 2.3.5 (Completeness Axiom)

If $A \subseteq \mathbb{R}$ is nonempty and bounded above, then A has a least upper bound in $\mathbb{R}$.

Example 2.3.6

Prove that if $a \in \mathbb{R}$ and $a \neq 0$, then $a^2 > 0$.

Proof

Since $a \neq 0$, the trichotomy law gives two cases.

Case 1: $a \in P$. By closure under multiplication,

$$a^2 = a \cdot a \in P, \quad \text{so } a^2 > 0.$$

Case 2: $a \notin P$. Since $a \neq 0$, trichotomy gives $-a \in P$, so by closure under multiplication $(-a)(-a) \in P$. It remains to show that $a^2 = (-a)(-a)$:

$$\begin{aligned} a + (-a) = 0 &\implies (-a)(a + (-a)) = (-a) \cdot 0 = 0 \\ &\implies (-a)(a) + (-a)(-a) = 0 \\ &\implies (-a)(-a) = -((-a)a) = -(-a^2) = a^2. \end{aligned}$$

Hence $a^2 = (-a)(-a) \in P$, so $a^2 > 0$. In either case $a^2 > 0$. ■

2.3.1 The Absolute Value**Definition 2.3.7**

Given $a \in \mathbb{R}$, we define the **absolute value** $|a|$ by

$$|a| = \begin{cases} a, & a \geq 0, \\ -a, & a < 0. \end{cases}$$

Theorem 2.3.8 (a) $|ab| = |a| |b|$, $\forall a, b \in \mathbb{R}$

(b) $|a|^2 = a^2$, $\forall a \in \mathbb{R}$

(c) $(\forall c > 0) [(|a| \leq c) \iff (-c \leq a \leq c)]$

(d) $(\forall a \in \mathbb{R}) -|a| \leq a \leq |a|$

You can prove all of them by cases.

Proposition 2.3.9 (Triangle Inequality)

If $a, b \in \mathbb{R}$, then $|a + b| \leq |a| + |b|$.

Proof

Let $a, b \in \mathbb{R}$. We know that

$$\begin{aligned} -|a| &\leq a \leq |a| \\ -|b| &\leq b \leq |b|. \end{aligned}$$

thus,

$$\begin{aligned} -|a| - |b| &\leq a + b \leq |a| + |b| \\ \iff -(|a| + |b|) &\leq a + b \leq |a| + |b| \end{aligned}$$

then we apply the absolute value theorem (c), we get

$$|a + b| \leq |a| + |b|.$$

■

Corollary 2.3.10 (i) $||a| - |b|| \leq |a - b|$.

(ii) $|a - b| \leq |a| + |b|$

Proof of (1)

Let $a, b \in \mathbb{R}$. We know

$$\begin{aligned} a &= a - b + b \implies |a| = |(a - b) + b| \leq |a - b| + |b| \\ &\implies |a| - |b| \leq |a - b| \end{aligned}$$

$$\begin{aligned} b &= b - a + a \implies |b| = |(b - a) + a| \leq |b - a| + |a| \\ &\implies |b| - |a| \leq |a - b|. \\ &\implies |a| - |b| \geq -|a - b| \end{aligned}$$

thus,

$$-|a - b| \leq |a| - |b| \leq |a - b|$$

Therefore,

$$||a| - |b|| \leq |a - b|$$

■

Proof of (2)

Let $a, b \in \mathbb{R}$. We know

$$|a - b| = |a + (-b)| \leq |a| + |-b| \leq |a| + |b|.$$

■

2.4 Bounds

This is an additional property of real numbers that is not satisfied by rational numbers.

Definition 2.4.1 (Bounded Sets)

Let S be a nonempty subset of $\mathbb{R}$. The set S is said to be

- (i) bounded above: there exists $u \in \mathbb{R}$ such that $x \leq u$ for all $x \in S$.
- (ii) bounded below: there exists $w \in \mathbb{R}$ such that $w \leq x$ for all $x \in S$.
- (iii) bounded: it is both bounded above and below.
- (iv) unbounded: not bounded.

If there exists an upper bound u_0 of S such that whenever u is an upper bound for S we have $u_0 \leq u$, then u_0 is called the least upper bound, or the supremum, of S .

$$u_0 := \sup S$$

If there exists a lower bound w_0 of S such that whenever w is a lower bound for S we have $w_0 \geq w$, then w_0 is called the greatest lower bound, or the infimum, of S .

$$w_0 := \inf S$$

Remark

There can be only one supremum/infimum of a given set S of $\mathbb{R}$.

Idea: Suppose u_1 and u_2 are both suprema of S .

Since u_1 is a supremum and u_2 is an upper bound, we get

$$u_1 \leq u_2.$$

Since u_2 is a supremum and u_1 is an upper bound, we get

$$u_2 \leq u_1.$$

Thus, $u_1 = u_2$.

Lemma 2.4.2

An upper bound u of a nonempty set S in $\mathbb{R}$ is the supremum of S if and only if

$$(\forall \epsilon > 0)(\exists x_\epsilon \in S) u - \epsilon < x_\epsilon.$$

Proof

To start off, we need to prove

$$u = \sup S \implies \forall \epsilon > 0, \exists x_\epsilon \in S, u - \epsilon < x_\epsilon.$$

Suppose $u = \sup S$.

Let $\epsilon > 0$, then

$$u - \epsilon < u,$$

which means $u - \epsilon$ is not an upper bound of S .

Thus, $\exists x_\epsilon \in S$ such that

$$u - \epsilon < x_\epsilon.$$

Now we prove the converse.

Suppose for every $\epsilon > 0$, $\exists x_\epsilon \in S$, $u - \epsilon < x_\epsilon$. Assume u is not the supremum; say $w = \sup S$. Then,

$$w < u \implies u - w > 0.$$

Take $\epsilon = u - w$. By hypothesis, $\exists x_\epsilon \in S$ such that

$$u - \epsilon < x_\epsilon \implies u - (u - w) < x_\epsilon \implies w < x_\epsilon.$$

But $x_\epsilon \in S$ and $w = \sup S$ means w is an upper bound for S , so $x_\epsilon \leq w$ — contradicting $w < x_\epsilon$. Therefore u must in fact be the supremum, i.e., $u = \sup S$. ■

Example 2.4.3

Consider the set $S = \{x \in \mathbb{R} : 0 \leq x \leq 1\}$.

– $\sup S = 1$.

Proof

Since $x < 1$ for all $x \in S$, 1 is an upper bound.

Say $u = \sup S$, so $0 \leq u \leq 1$. Suppose $u \neq 1$; then

$$0 \leq u < 1 \implies 1 - u > 0.$$

Take $\epsilon = \frac{1-u}{2}$. Thus,

$$u < u + \epsilon \implies u < u + \frac{1-u}{2} = \frac{u+1}{2} < 1,$$

so $\frac{u+1}{2} \in S$. But u is an upper bound for S , so it should satisfy $\frac{u+1}{2} \leq u$ — yet we just showed $u < \frac{u+1}{2}$, a contradiction. ■

– $\inf S = 0$.

Proof

Since $x \geq 0$ for all $x \in S$, 0 is a lower bound. Assume $w = \inf S \neq 0$.

Then

$$w > 0.$$

Since w is a lower bound for S and $0 \in S$, we must have $w \leq 0$, contradicting $w > 0$. ■

2.5 The Completeness Property of $\mathbb{R}$

Axiom (Completeness Property)

Every nonempty set of real numbers that has an upper bound also has a supremum in $\mathbb{R}$.

Corollary 2.5.1

Every nonempty set of real numbers that has a lower bound also has an infimum in $\mathbb{R}$.

Proof

Let $S \subseteq \mathbb{R}$ be nonempty with a lower bound w , that is, $w \leq s$ for all $s \in S$. As a consequence, $-w \geq -s$ for all $s \in S$.

Define $S' = \{-s : s \in S\}$. Then S' is nonempty, and

$$-w \geq -s, \forall s \in S \implies -w \text{ is an upper bound for } S'.$$

By the Completeness Axiom, S' has a supremum. Say

$$u := \sup S' \in \mathbb{R}.$$

We claim that $-u = \inf S$.

- **Lower bound:** Since $u = \sup S' \geq -s$ for all $s \in S$, we get $-u \leq s$ for all $s \in S$. So $-u$ is a lower bound for S .
- **Greatest lower bound:** Let w be any lower bound for S . As shown above, $-w$ is then an upper bound for S' , so $u = \sup S' \leq -w$, which gives $w \leq -u$.

Thus $-u$ is a lower bound for S that is at least as large as every other lower bound, so $-u = \inf S$. ■

Example 2.5.2

$$S = \left\{ \left(1 + \frac{1}{n}\right)^n : n \in \mathbb{N} \right\}$$

You can try $1 < \left(1 + \frac{1}{n}\right)^n < 3$.

2.6 The Archimedean Property

We now prove that $\mathbb{N}$ is not bounded above in $\mathbb{R}$. The proof combines two ingredients:

- Completeness of $\mathbb{R}$.
- The inductive property of $\mathbb{N}$ (that $\mathbb{N}$ is closed under $+1$).

Definition 2.6.1 (Archimedean Property)

If $x \in \mathbb{R}$, then there exists $n_x \in \mathbb{N}$ such that $x \leq n_x$.

Proof

Suppose, for contradiction, that $\exists x \in \mathbb{R}$ such that $x > n$ for all $n \in \mathbb{N}$.

Then x is an upper bound for $\mathbb{N}$. Since $1 \in \mathbb{N}$, $\mathbb{N} \neq \emptyset$, so by the Completeness Axiom, $\mathbb{N}$ has a supremum in $\mathbb{R}$.

Say $u = \sup \mathbb{N}$. Since

$$u - 1 < u,$$

$u - 1$ is not an upper bound for $\mathbb{N}$. Hence

$$\exists m \in \mathbb{N} \text{ such that } u - 1 < m,$$

which gives

$$u < m + 1.$$

But $m + 1 \in \mathbb{N}$ (since $\mathbb{N}$ is closed under $+1$), so $m + 1 \leq u$, contradicting $u < m + 1$.

Therefore no such x exists; that is, for every $x \in \mathbb{R}$ there exists $n_x \in \mathbb{N}$ with $x \leq n_x$. ■

Proposition 2.6.2

There exists a real number x such that $x^2 = 2$.

Proof

Define

$$S = \{t \in \mathbb{R} : t \geq 0, t^2 < 2\}.$$

We have $1 \in S$, so $S \neq \emptyset$, and 2 is an upper bound for S (if $t > 2$ then $t^2 > 4 > 2$, so $t \notin S$). By the Completeness Axiom, S has a supremum in $\mathbb{R}$.

Let $x = \sup S$. Note $x \geq 1 > 0$. We show $x^2 = 2$ by ruling out $x^2 < 2$ and $x^2 > 2$.

Case 1: $x^2 < 2$. We find $n \in \mathbb{N}$ large enough that $x + \frac{1}{n} \in S$, contradicting that x is an upper bound for S . Since $x \geq 1$,

$$\left(x + \frac{1}{n}\right)^2 = x^2 + \frac{2x}{n} + \frac{1}{n^2} \leq x^2 + \frac{2x}{n} + \frac{1}{n} = x^2 + \frac{2x+1}{n}.$$

Since $x^2 < 2$, choose $n \in \mathbb{N}$ (by the Archimedean Property) large enough that

$$\frac{2x+1}{n} < 2 - x^2.$$

Then

$$\left(x + \frac{1}{n}\right)^2 < x^2 + (2 - x^2) = 2,$$

so $x + \frac{1}{n} \in S$. But $x + \frac{1}{n} > x = \sup S$, a contradiction. Hence $x^2 \geq 2$.

Case 2: $x^2 > 2$. We find $n \in \mathbb{N}$ large enough that $x - \frac{1}{n}$ is an upper bound

for S smaller than x , contradicting that $x = \sup S$. Since

$$\left(x - \frac{1}{n}\right)^2 = x^2 - \frac{2x}{n} + \frac{1}{n^2} > x^2 - \frac{2x}{n},$$

choose $n \in \mathbb{N}$ large enough that

$$\frac{2x}{n} < x^2 - 2.$$

Then

$$\left(x - \frac{1}{n}\right)^2 > x^2 - (x^2 - 2) = 2.$$

So for any $t \in S$, we have $t^2 < 2 < \left(x - \frac{1}{n}\right)^2$. Since both $t \geq 0$ and $x - \frac{1}{n} \geq 0$ (as $x \geq 1$ and n is large), taking square roots of $t^2 < \left(x - \frac{1}{n}\right)^2$ gives $t < x - \frac{1}{n}$. Thus $x - \frac{1}{n}$ is an upper bound for S smaller than $x = \sup S$, a contradiction. Hence $x^2 \leq 2$.

Combining both cases, $x^2 = 2$. ■